Windows Server Patching Process – End-to-End SOP

1. Patch Governance and Approval Process

Overview

This page describes the end-to-end patching lifecycle across all environments from test server validation through to production deployment and application sign-off. The process is structured into four sequential phases, each with defined timelines, responsibilities, and escalation paths.


Phase 1
Test Env Patching
Phase 2
Non-Prod Patching
Phase 3
Prod Deployment
Phase 4
App Validation
Step 1
Apply N-1 patch on test servers
Step 2
Verify — no issues observed
Step 3
Raise CRQ for Non-Prod patching
Step 1 (–4 wks)
Send patching notice; update tester details and schedules
Escalation (–3 wks)
No tester details → escalate to SPOC / product owners
Step 2 (–2 wks)
Final reminder sent to testers
Step 3 — patch day
Patching runs per schedule
Step 4
Notify completion
Step 5
Testers validate within 48 hrs; rollback if needed
Step 1 (–4 wks)
Publish patch list; create JIRA stories; share schedule
Escalation (–3 wks)
No tester details → escalate to business partners / owners
Step 2 (–1 wk)
Final list + reminder to all stakeholders
Step 3 — locked
Patches scheduled; no rollback possible
Step 4
Notify completion
Step 1
Patching starts; Bridge details shared with testers to join
Step 2
Issues noticed → rollback / restore backup
Step 3
Update patching workbooks
Escalation / rollback path
No rollback once started

Patch Approval Process :

The monthly patching process begins with the publication of a list of patches on the Confluence page for the respective month. Once the patch list is available, representatives from the Security, Windows, and Engineering teams review each patch and provide their approval or feedback on the Confluence page.

After the review process is completed, patches that receive approval from the majority of the stakeholders are approved in SCCM. These approved patches are then added to the appropriate Software Update Group (SUG) and are deployed to the target servers during the scheduled patching window. The details are available on the page Monthly Patch Approvals

Patching Process Workflow:

Phase 1 — Test Environment Patching

Objective: Validate the patch package on test servers before promoting to Non-Prod or Production.

Steps:

1. Apply N-1 Patch on Test Servers The patch to be deployed (N-1, i.e., one version before the latest) is applied on designated test servers. This ensures the patch is exercised in an environment that mirrors the production configuration before any wider rollout.

2. Verify Stability — No Issues Observed. The infrastructure and application teams monitor the test servers for a defined observation period after patching. If no functional issues, service disruptions, or performance anomalies are identified, the phase is considered successful.

3. Raise a Change Request (CRQ) for Non-Prod. Once test validation passes, a formal Change Request is raised in the ITSM tool (ServiceNow / JIRA) to authorize patching of the Non-Production environment. The CRQ must reference test results and attach the approved patch list.

Phase 2 — Non-Prod Patching

Objective: Patch all non-production servers on a controlled schedule with tester validation as the gate before production.

Timeline and Steps:

Timeline	Action
–4 weeks before patch date	Patching notification sent to all stakeholders. Testers are asked to confirm/update their tester details and validate schedules.
–3 weeks (escalation)	If tester details have not been published or confirmed, the matter is escalated to the Patching SPOC, Product Owner, or Application Owner for resolution.
–2 weeks	Final reminder sent to testers. Validation assignments and bridge details are confirmed.
Patch date	Patching commences per the approved schedule. If a tester does not appear during the patching window, patching still proceeds as scheduled.
Post-patch	Completion notification is sent to all stakeholders.
Within 48 hours of completion	Testers validate application functionality and either confirm success or raise a rollback request.

> ***Key Note:*** *Patching proceeds on schedule regardless of tester attendance. Tester no-show does not block the patching window.*

Phase 3 — Production Deployment

Objective: Deploy approved patches to the production environment following a rigorously communicated schedule. Rollback is not available once deployment begins.

Timeline and Steps:

Timeline	Action
–4 weeks before patch date	The final patching list is published. JIRA stories are created for each server or server group. Schedules are shared with testers and business partners.
–3 weeks (escalation)	If tester or business partner details are missing from the published list, an escalation is raised to the Business Partners or Application Owners. This must be resolved before the–1–week milestone.
–1 week	Final patching list republished with a final reminder to testers and business partners. No further changes to the scope are accepted after this point.

Patch date	Patches are scheduled for deployment. **At this stage, rollback is not possible.** All teams must be on standby.
Post-patch	Deployment completion notification is sent to all stakeholders and business partners.

> ***Critical:*** *Once production patching begins, rollback is not supported. All validation gates must be completed in Phases 1 and 2 before production deployment proceeds.*

Phase 4 — Application Validation

Objective: Confirm that all production-patched systems are functioning correctly. Identify and remediate issues immediately post-deployment.

Steps:

1. Patching Commences — Bridge Call Active As production patching begins, a Bridge call is opened and the details are shared with all assigned testers. Testers are expected to join at their designated validation slot to confirm application health in real time.

2. Issue Identified — Rollback or Restore from Backup If any issues are noticed during or immediately after patching, the responsible team initiates a rollback or restores from the most recent backup. Rollback decisions must be made within the defined post-patching validation window. Any rollback must be documented in the incident record.

3. Update Patching Workbooks Once validation is complete (pass or rollback), the patching workbooks are updated to reflect the final status of each server — patched successfully, rolled back, or deferred. This record feeds into the next patching cycle's baseline.

2. Patch Deployment and Implementation Process

Patch Approval Process: Created software update group SCCM

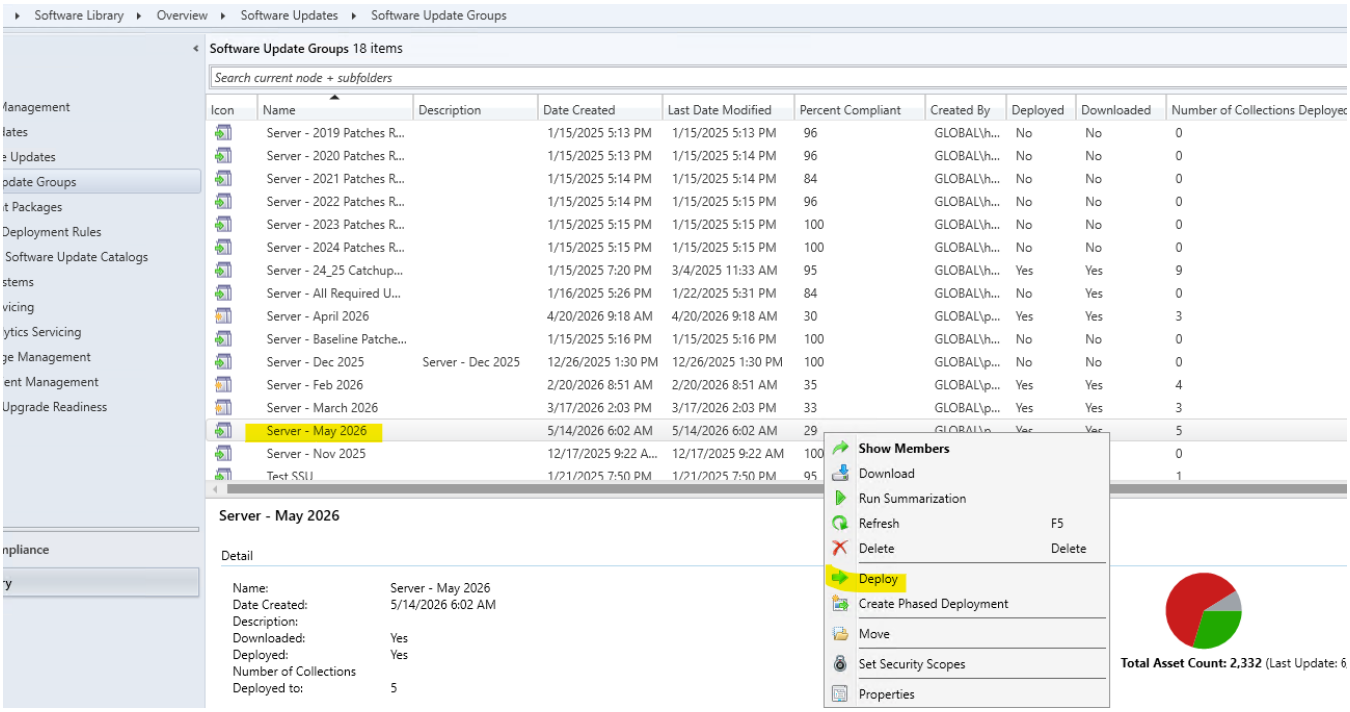
T1 Server Patch Deployment Process

Patch testing in Testing servers

Login to ORDPGBLSCCM11.global.cdk.com
Open **SCCM Console** and navigate to **Software Library > Software Updates > Software Update Groups**

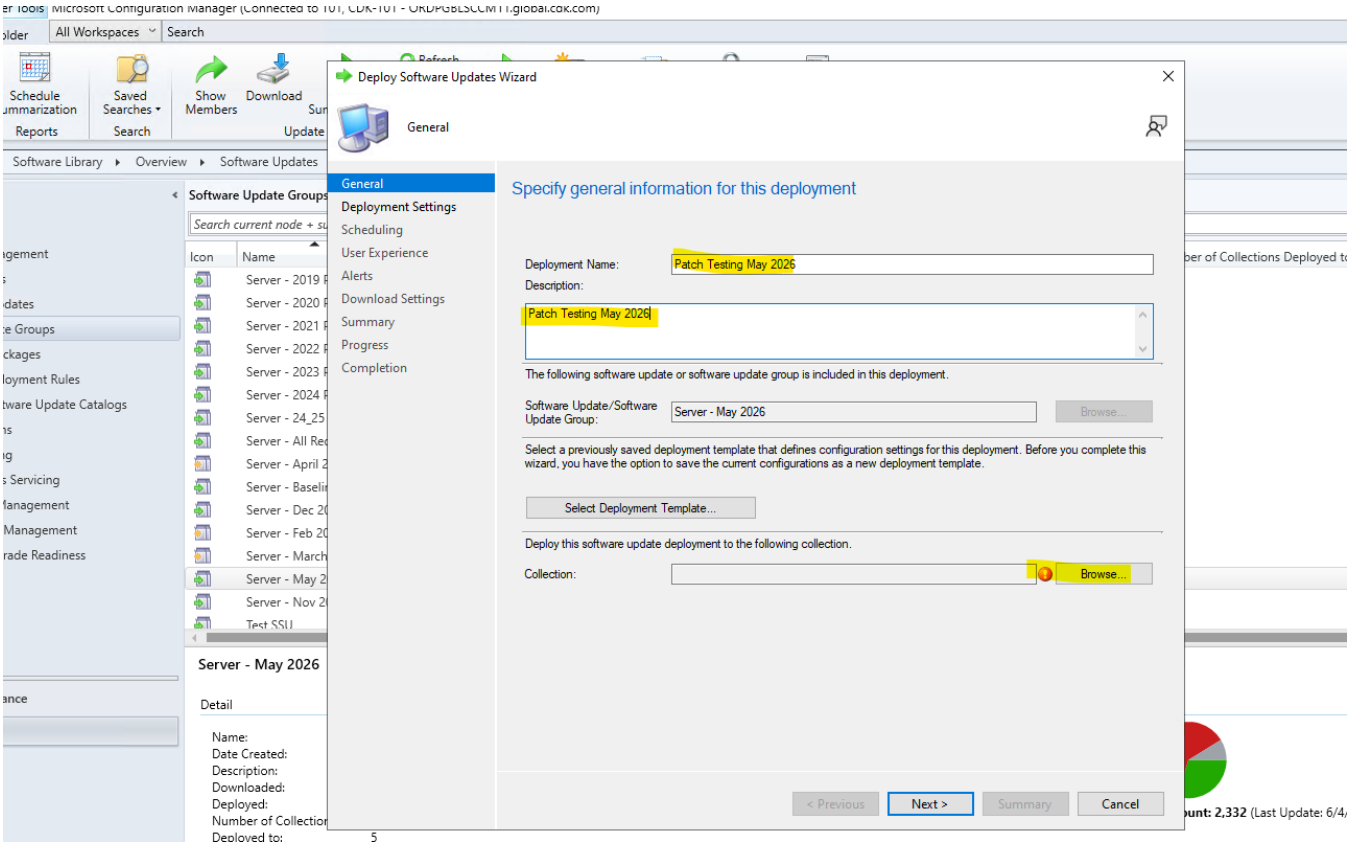
Select the required **Software Update Group (SUG)**., like "**Server - May 2026**"

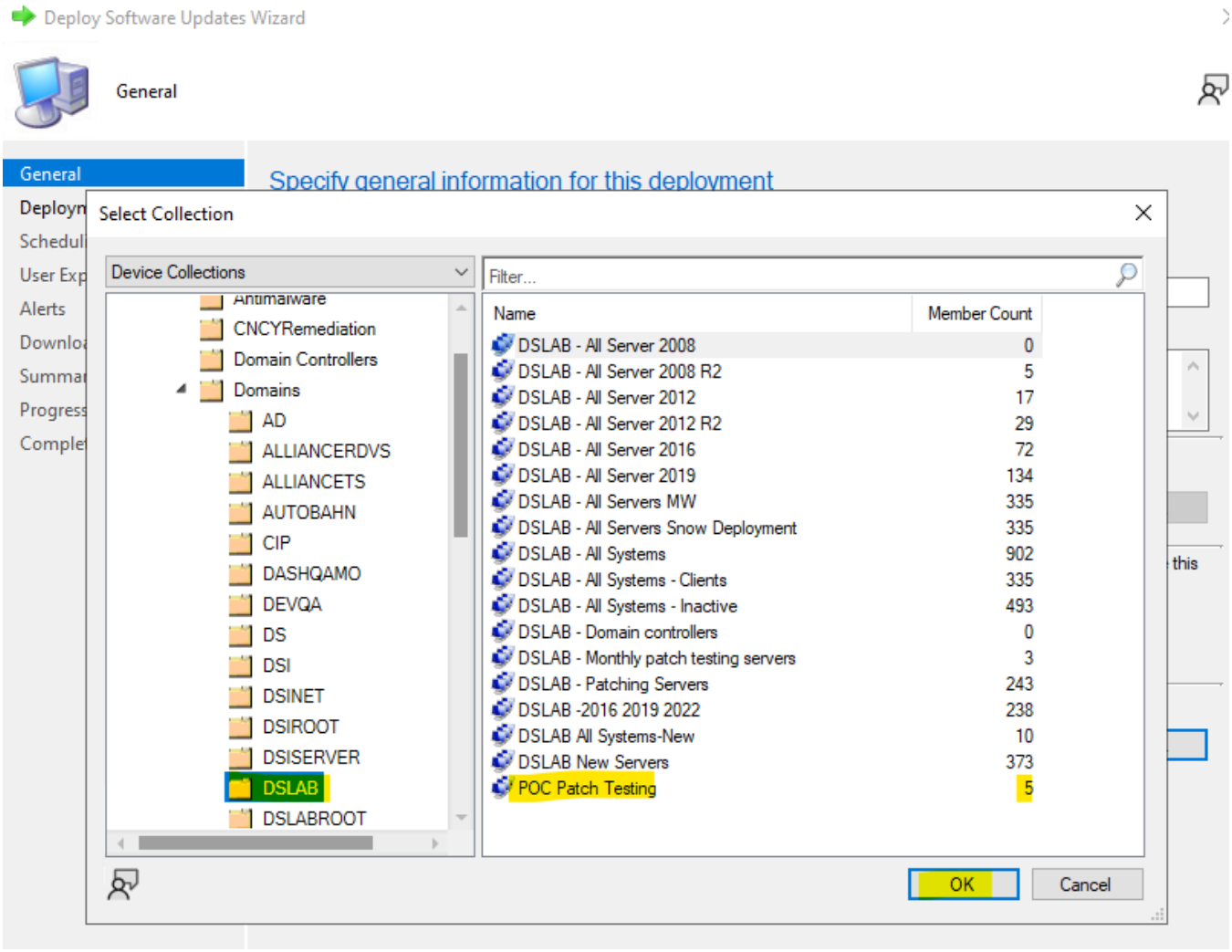
Right-click the SUG and select **Deploy**



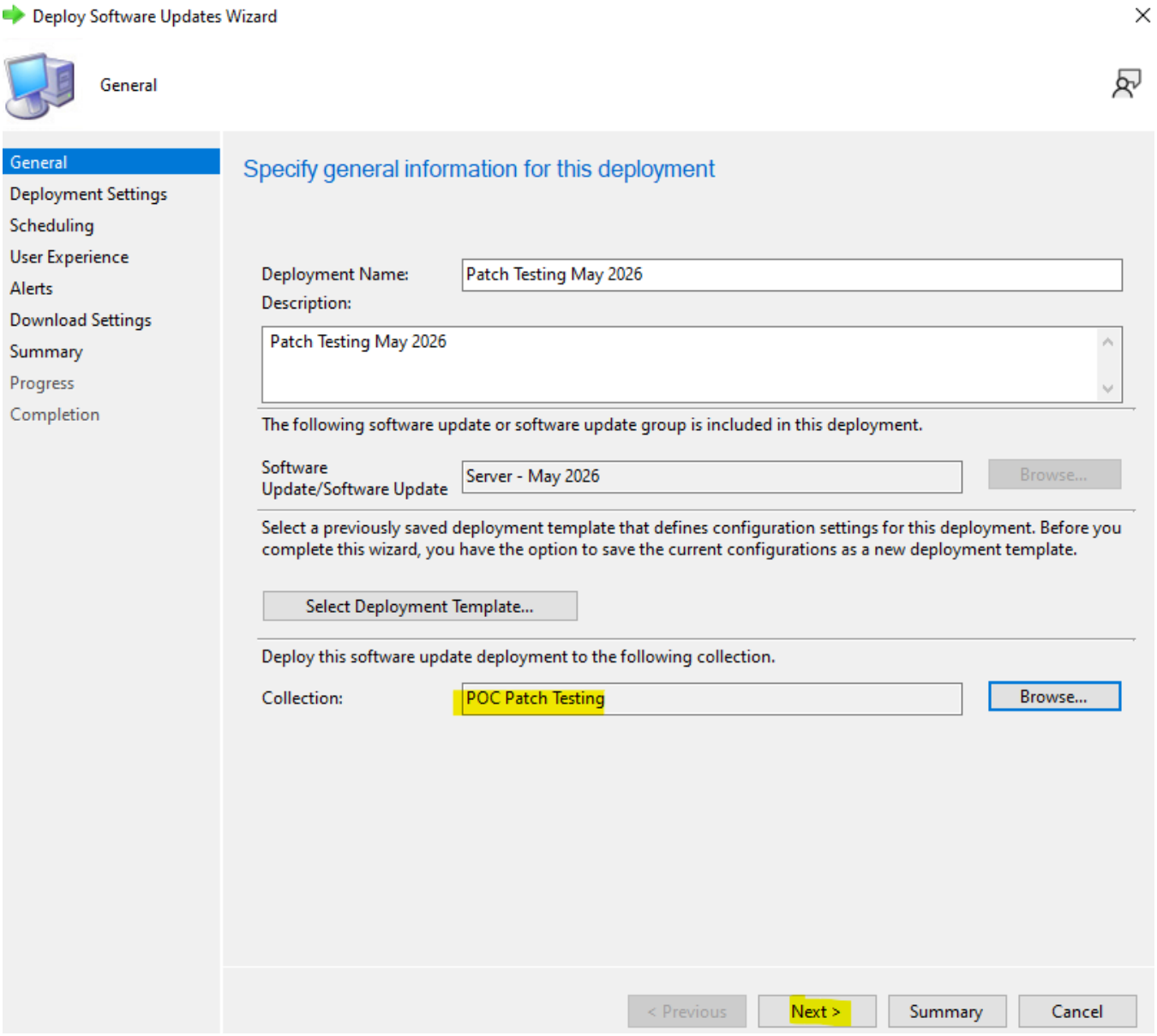
Update the Deployment as per requirement Like " **Patch Testing May 2026**"

Click **Browse** and select the target **Server Collection - "*POC Patch Testing*"**



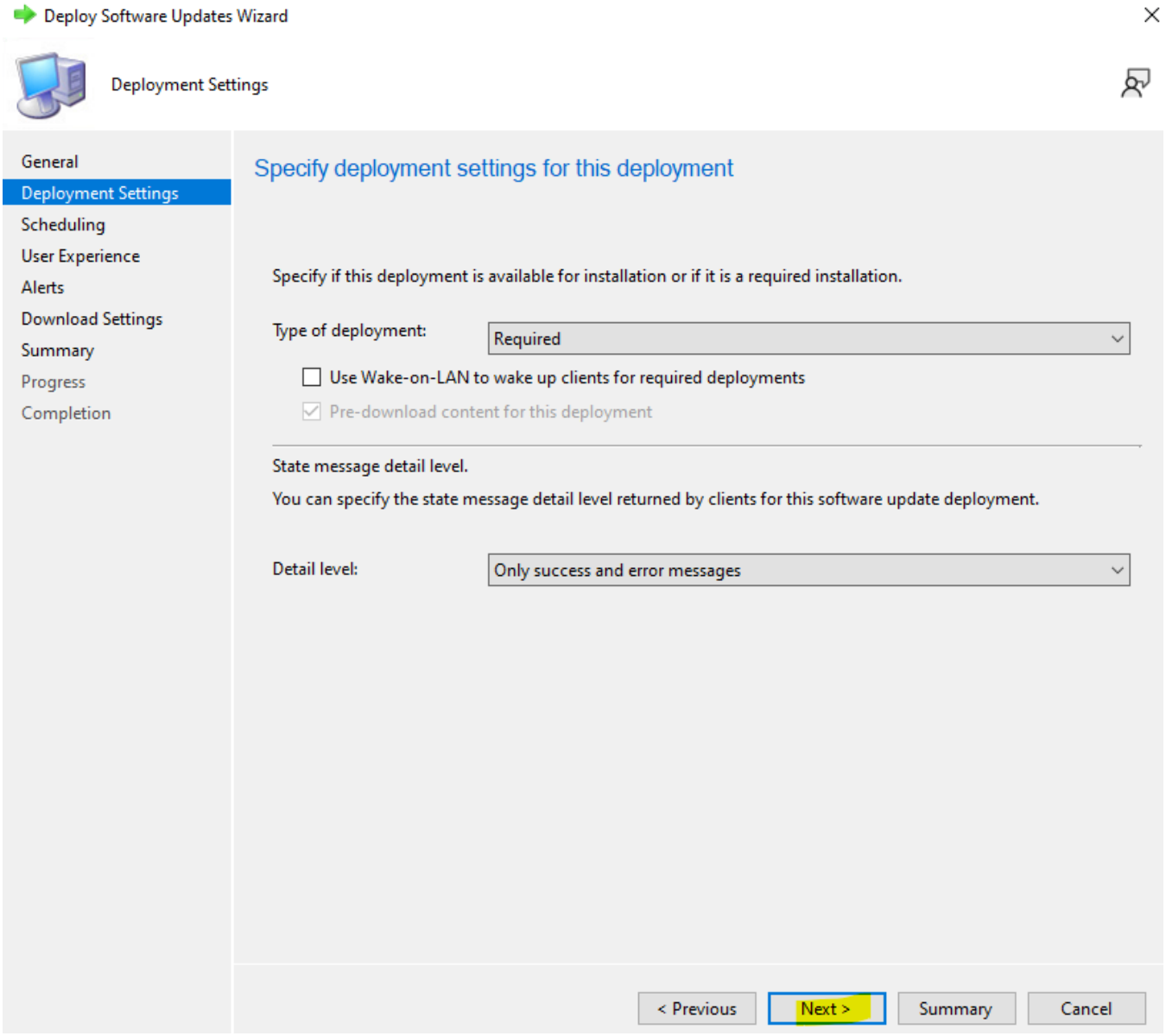

Deploy Software Updates Wizard
General
General
Specify general information for this deployment
Select Collection
Device Collections
Filter...
Antimalware
CNCYRemediation
Domain Controllers
Domains
AD
ALLIANCERDVS
ALLIANCETS
AUTOBAHN
CIP
DASHQAMO
DEVQA
DS
DSI
DSINET
DSIROOT
DSISERVER
DSLAB
DSLABROOT
Name
Member Count
DSLAB - All Server 2008 0
DSLAB - All Server 2008 R2 5
DSLAB - All Server 2012 17
DSLAB - All Server 2012 R2 29
DSLAB - All Server 2016 72
DSLAB - All Server 2019 134
DSLAB - All Servers MW 335
DSLAB - All Servers Snow Deployment 335
DSLAB - All Systems 902
DSLAB - All Systems - Clients 335
DSLAB - All Systems - Inactive 493
DSLAB - Domain controllers 0
DSLAB - Monthly patch testing servers 3
DSLAB - Patching Servers 243
DSLAB -2016 2019 2022 238
DSLAB All Systems-New 10
DSLAB New Servers 373
POC Patch Testing 5
OK
Cancel

Click **Next**



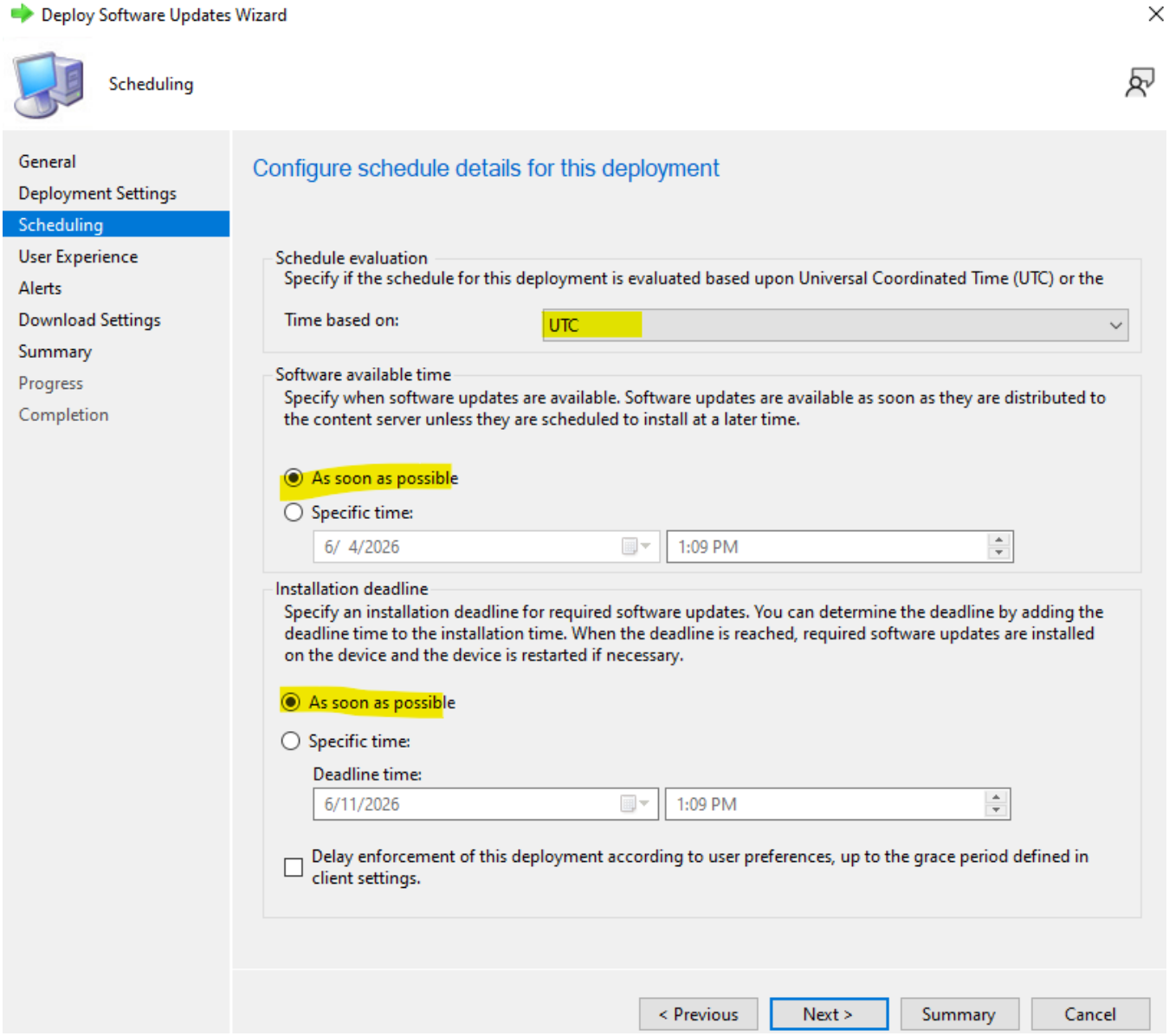
Keep the default setting : **Deployment Type** to **Required** and Detail Level: **Only Success and error messages**

Click **Next**



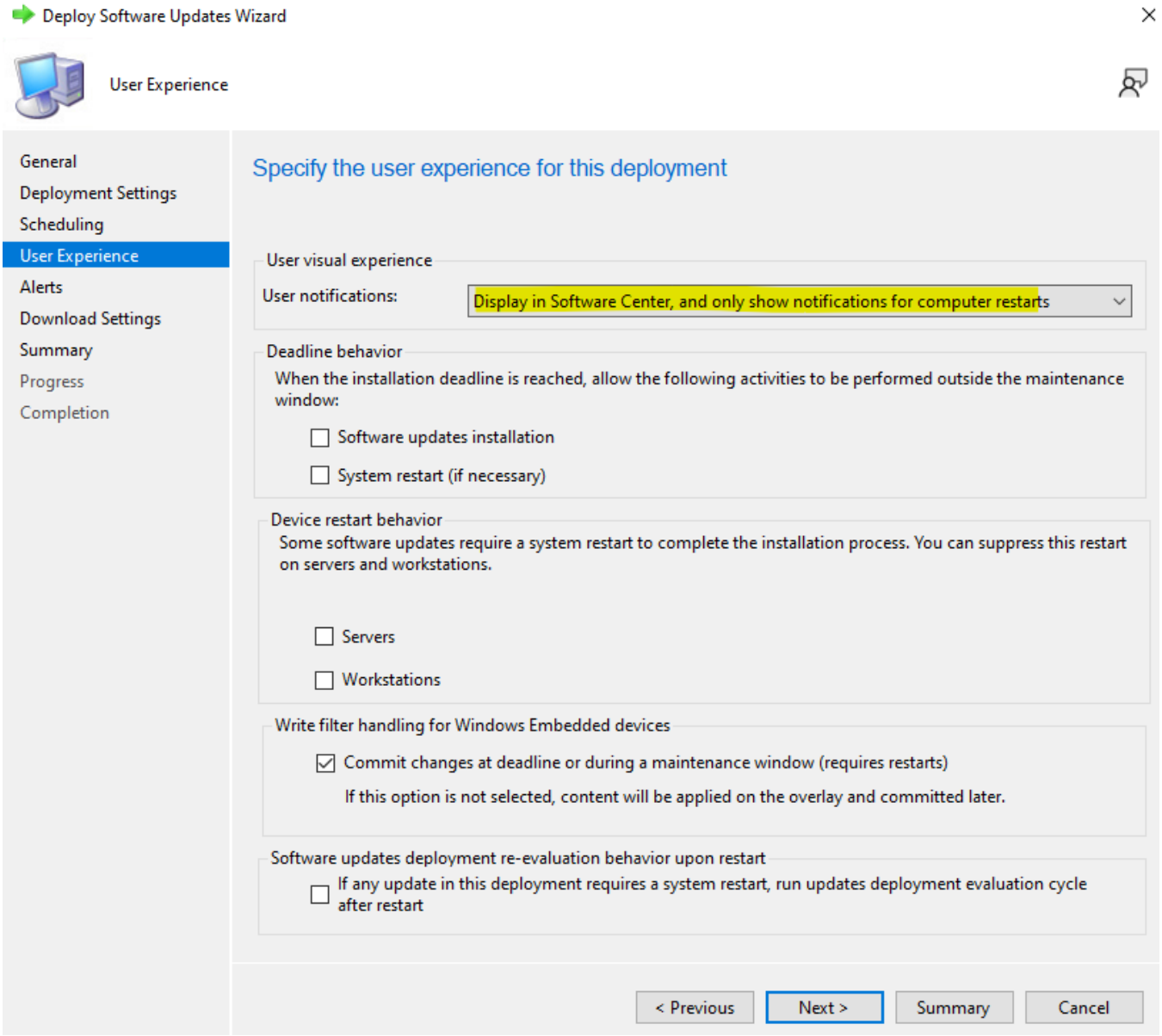
Enable **Use UTC**

Software available time to **As soon as possible &** Installation Deadline **As soon as possible**



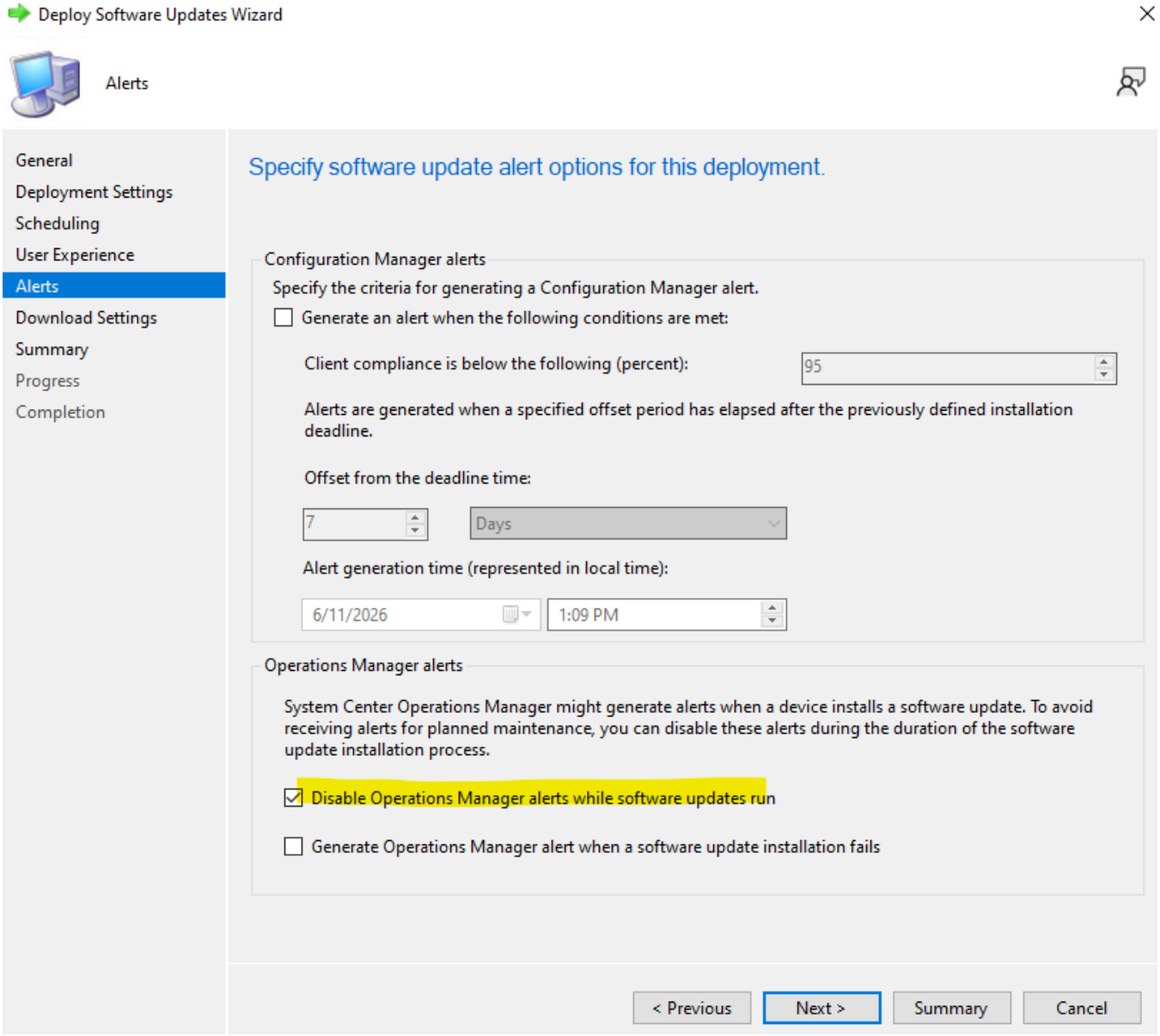
Change the User notifications: **Display in Software Center and only show notifications for computer restarts**

Click **Next**

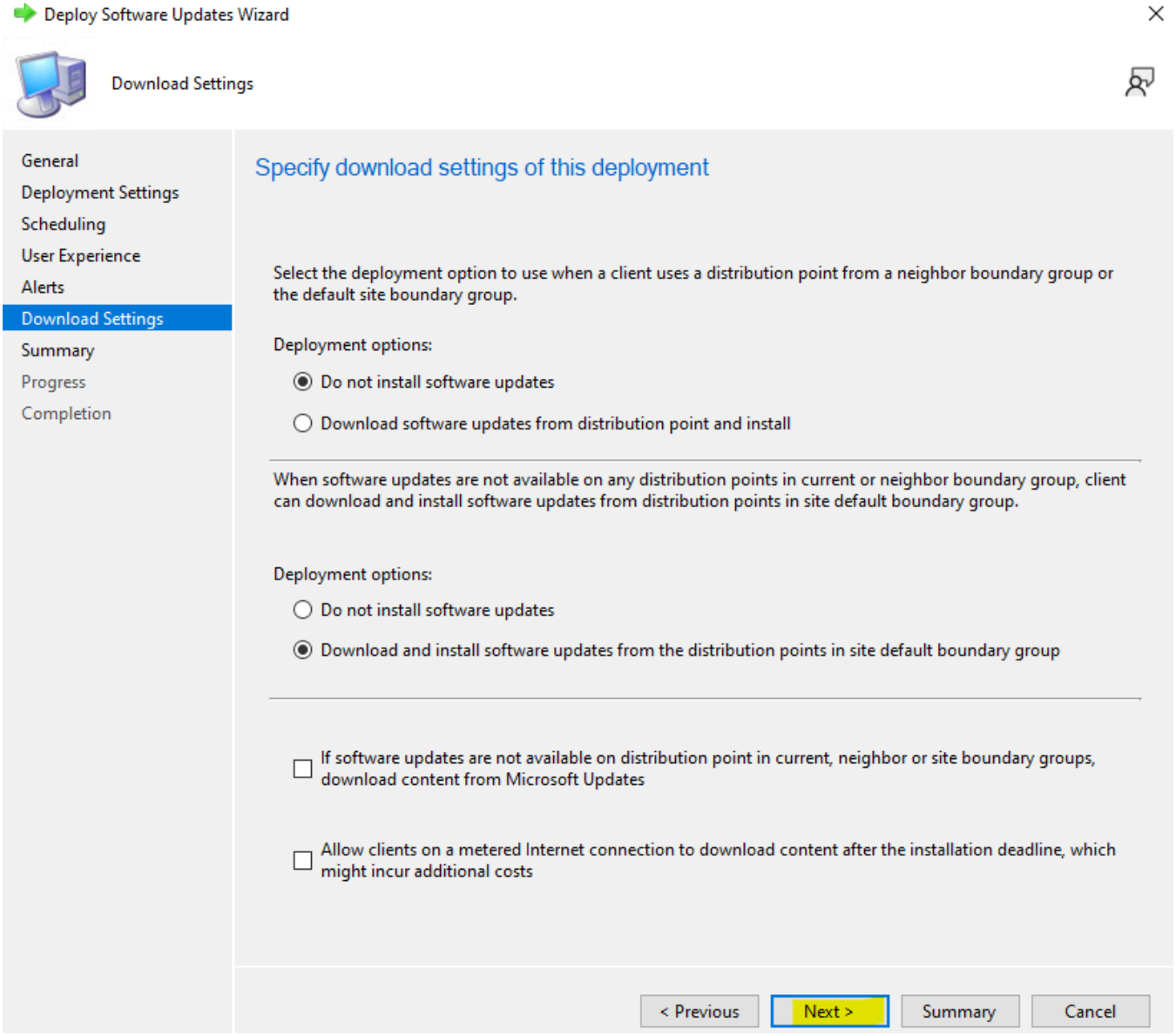


Enable **Disable Operations Manager alerts while software updates run**

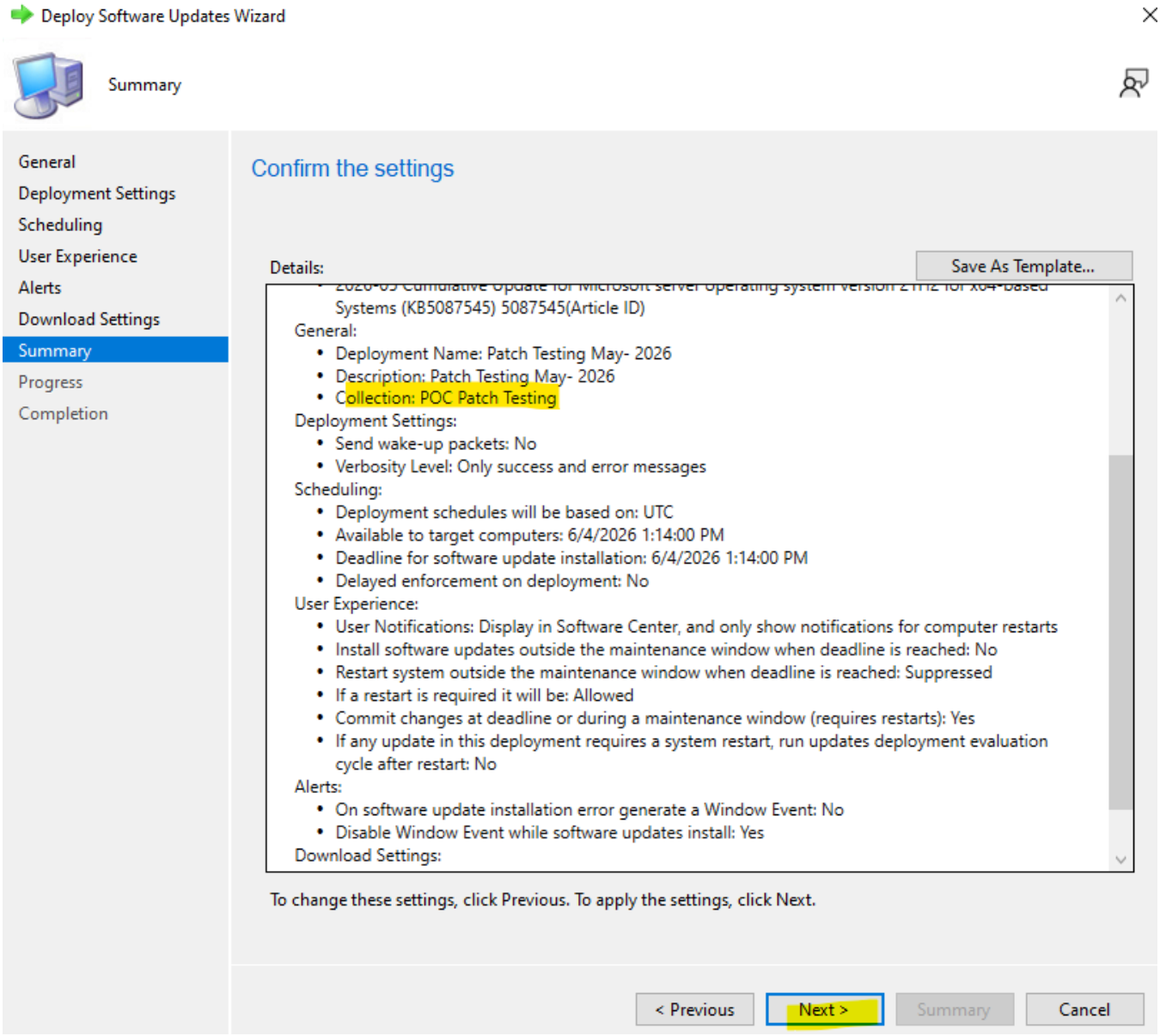
Click **Next**



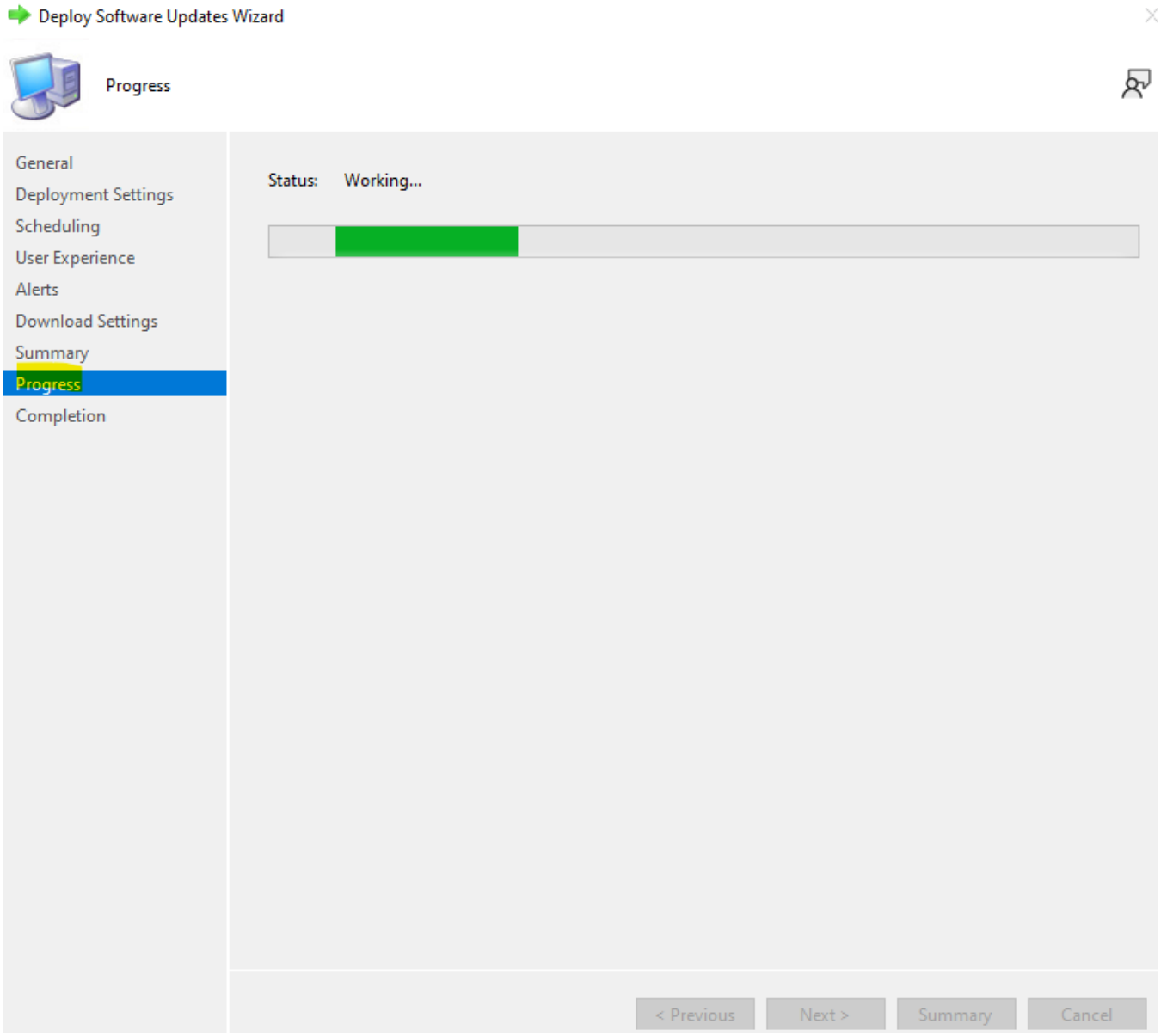
Keep the default setting and click **Next**

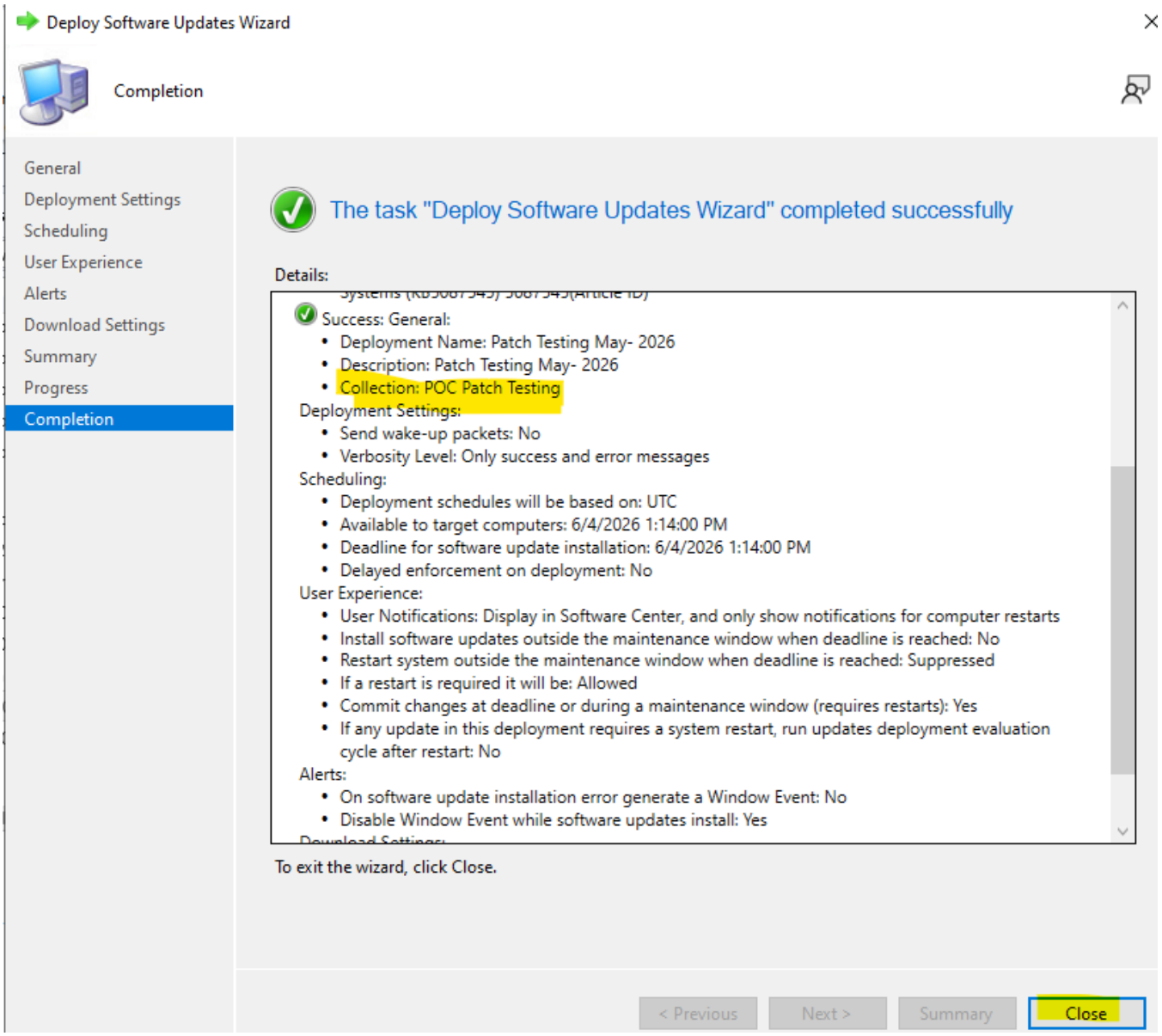

Deploy Software Updates Wizard
Download Settings
General
Deployment Settings
Scheduling
User Experience
Alerts
Download Settings
Summary
Progress
Completion
Specify download settings of this deployment
Select the deployment option to use when a client uses a distribution point from a neighbor boundary group or the default site boundary group.
Deployment options:
Do not install software updates
Download software updates from distribution point and install
When software updates are not available on any distribution points in current or neighbor boundary group, client can download and install software updates from distribution points in site default boundary group.
Deployment options:
Do not install software updates
Download and install software updates from the distribution points in site default boundary group
If software updates are not available on distribution point in current, neighbor or site boundary groups, download content from Microsoft Updates
Allow clients on a metered Internet connection to download content after the installation deadline, which might incur additional costs
< Previous
Next >
Summary
Cancel

click **Next**



Monitor deployment status for **Success**


Deploy Software Updates Wizard
Progress
General
Deployment Settings
Scheduling
User Experience
Alerts
Download Settings
Summary
Progress
Completion
Status: Working...
< Previous
Next >
Summary
Cancel

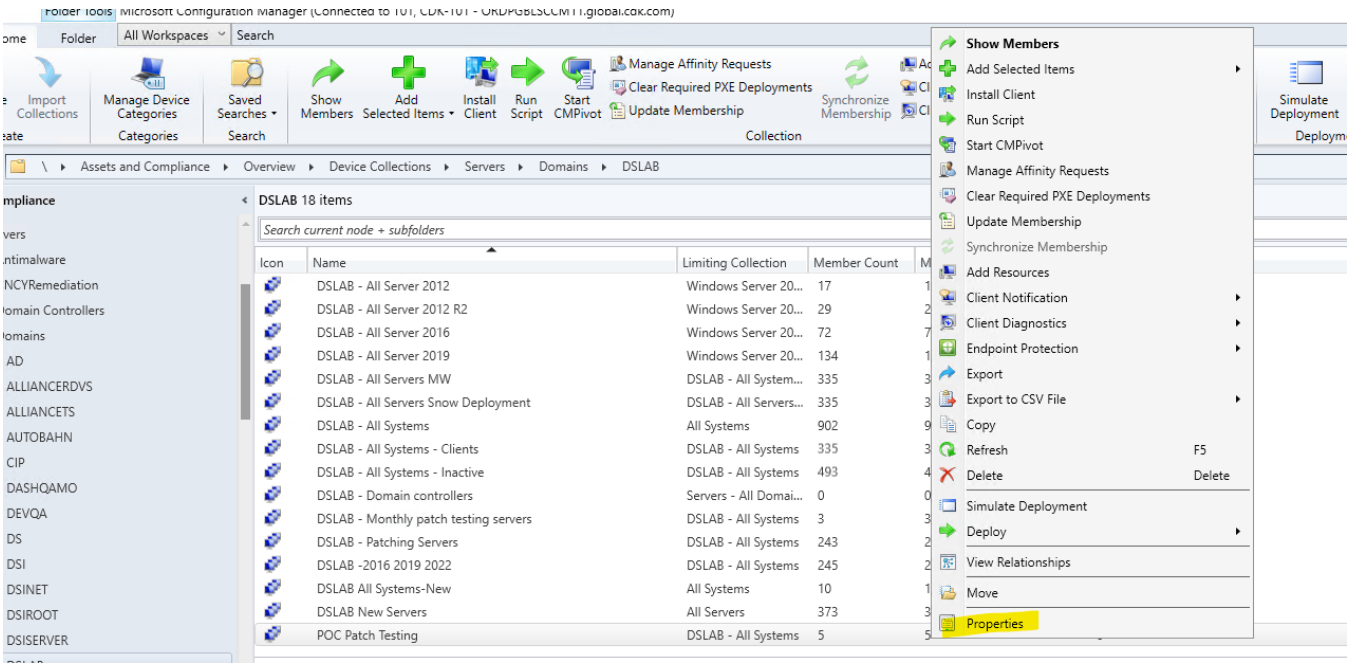


Setting MW on the Patch Testing collection:

Open **SCCM Console** and navigate to \Assets and Compliance\Overview\Device Collections\Servers\Domains\DSLAB

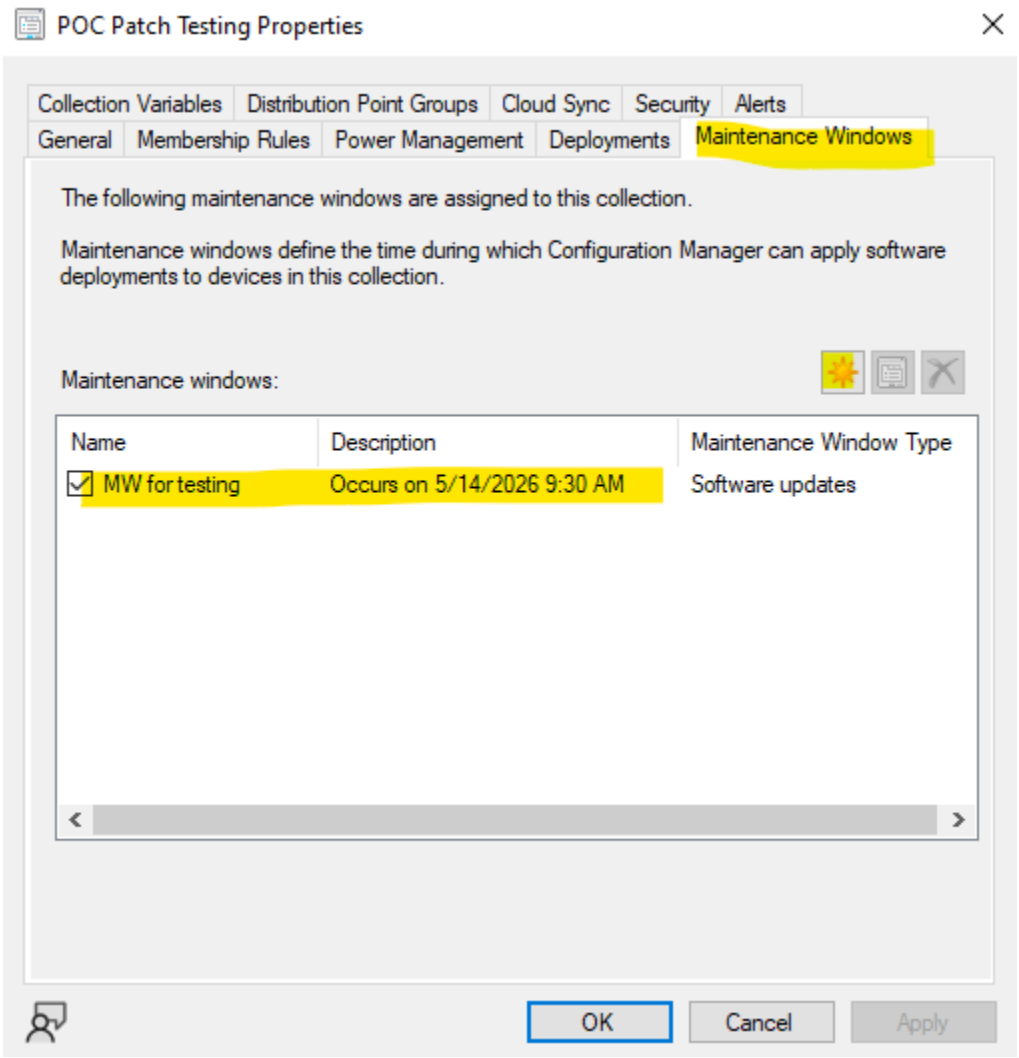
Locate and select the **Patch Testing Collection - POC Patch Testing**

Right-click the collection and select **Properties**.



Select the **Maintenance Windows** tab.

Click **Edit the existing MW**



Enter the **Maintenance Window Name** (e.g., *MW for Testing*).

Set the **Effective Date**.

Configure the **Start Time**. - **5 AM**

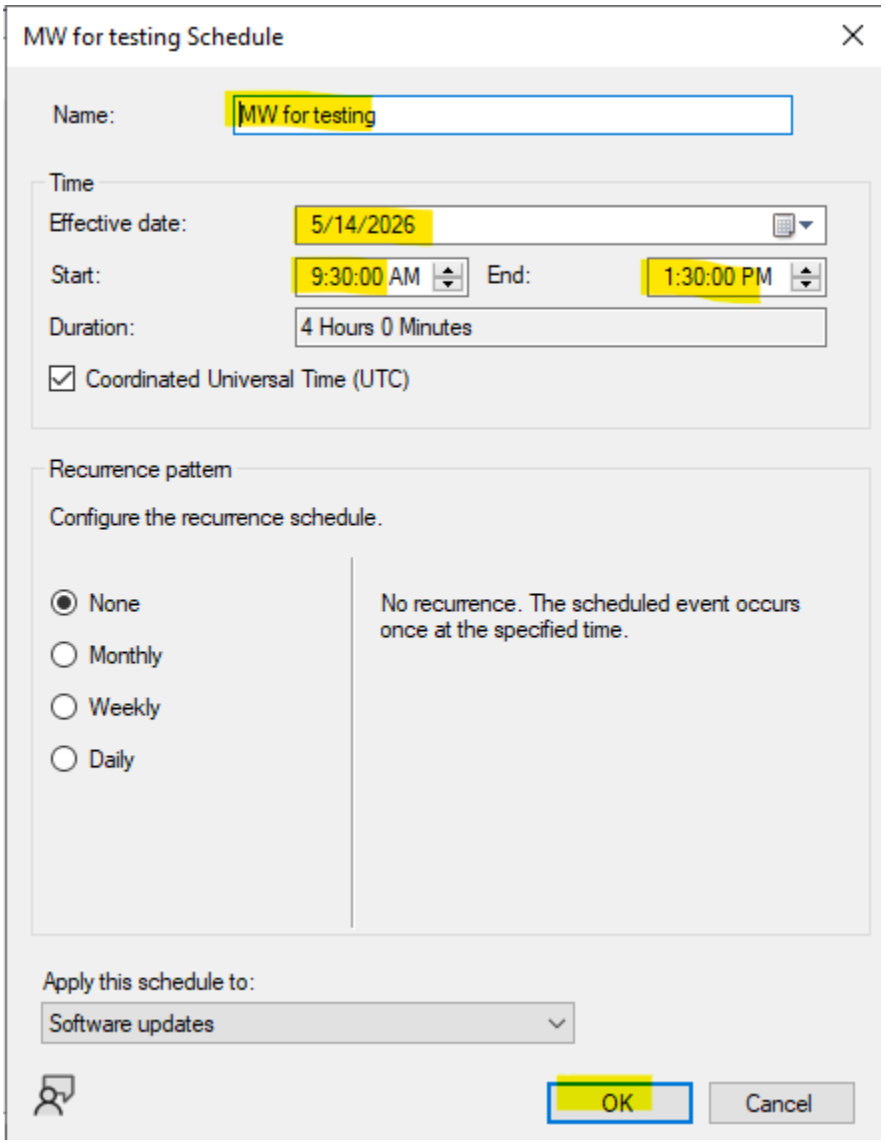
Configure the **End Time - 10 AM**

Select **Coordinated Universal Time (UTC)** if required.

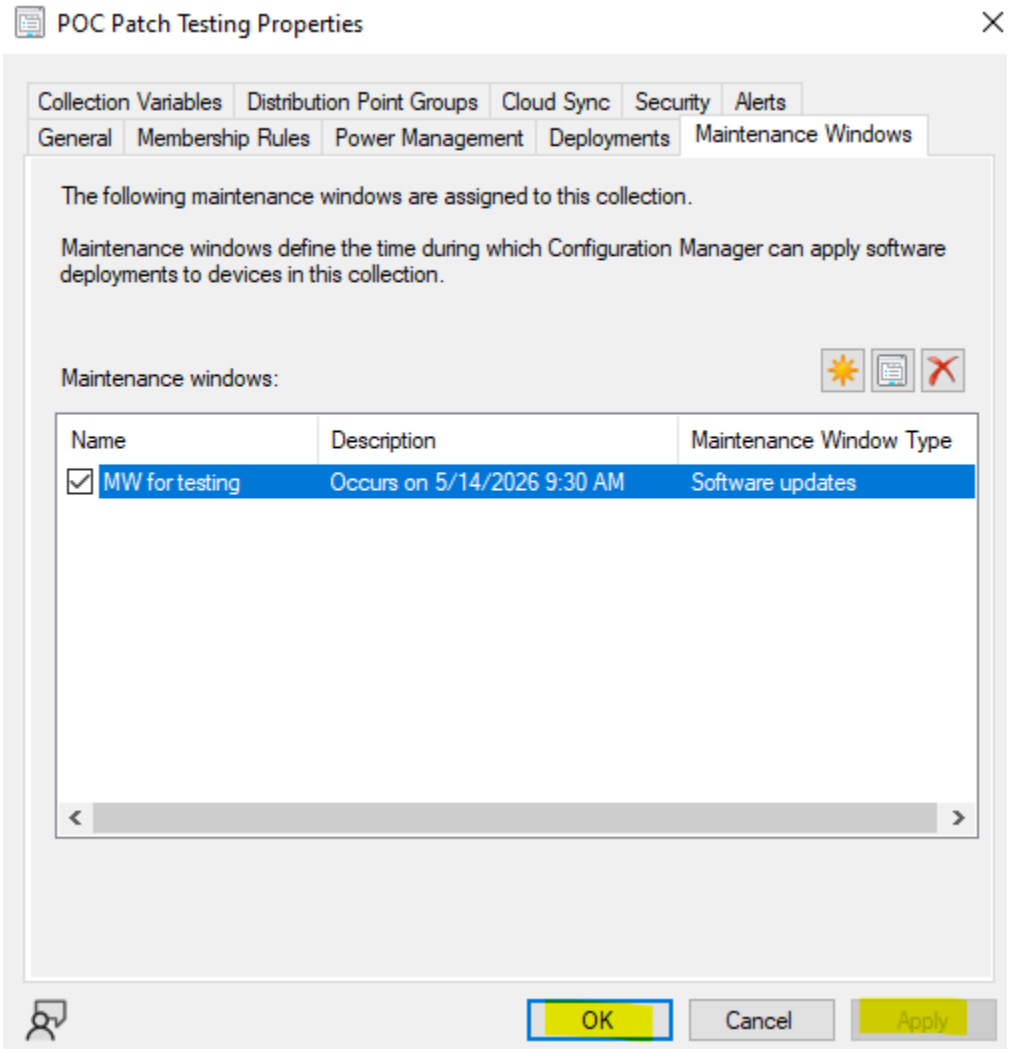
Under **Recurrence Pattern**, select **None** for a one-time maintenance window

Select **Apply this maintenance window to: Software Updates**.

Click **OK**



Click **Apply** and **OK** to save the collection properties



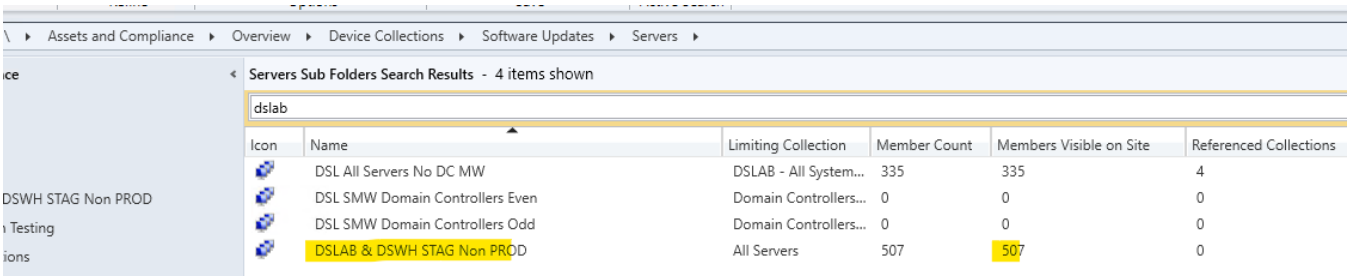
Non-PROD Server Patching:

Servers Detaills available in location:

Go to the location \Assets and Compliance\Overview\Device Collections\Software Updates\Servers

Collection Name : **DSLAB & DSWH STAG Non PROD**

The patch deployment process will be the same as mentioned in the patch deployment on test servers



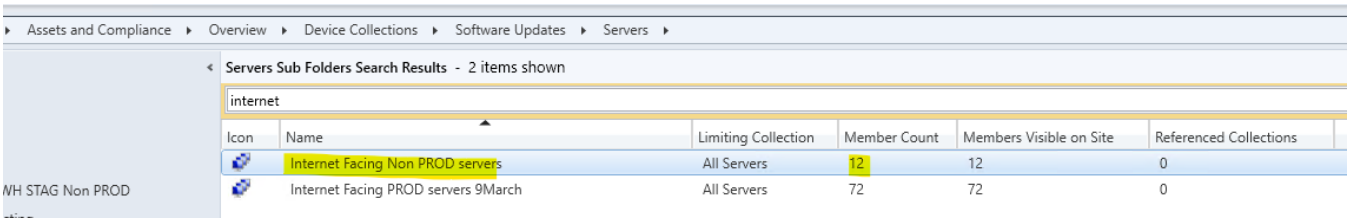
Internet-Facing Non-PROD Server Patching:

Server details available in the location:

Go to the location \Assets and Compliance\Overview\Device Collections\Software Updates\Servers

Collection Name : **Internet Facing Non PROD servers**

The patch deployment process will be the same as mentioned in the patch deployment on test servers



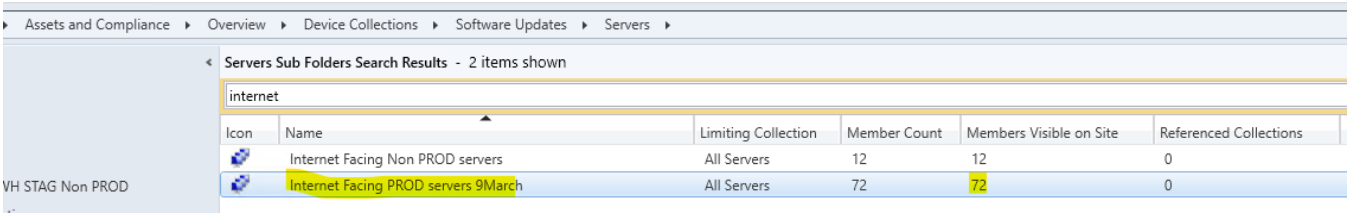
Internet-Facing PROD Server Patching:

Server details available in the location:

Go to the location \Assets and Compliance\Overview\Device Collections\Software Updates\Servers

Collection Name : **Internet Facing PROD servers 9March**

The patch deployment process will be the same as mentioned in the patch deployment on test servers



PROD Server Patching:

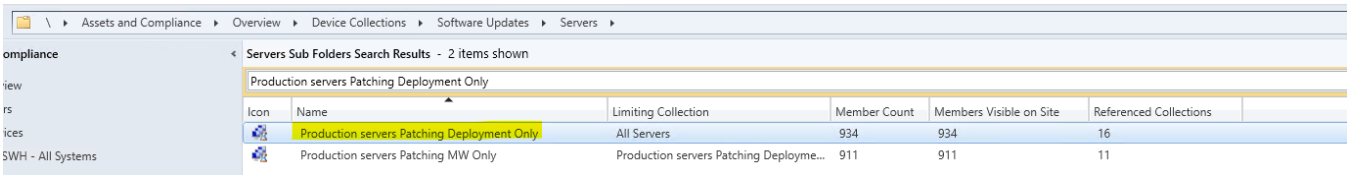
Patch Deployment Only: Since the PROD servers contain Cluster servers which need to be patched manually, this collection is only to deploy the Patches, ***Not to set any MW on this Collection***

Server details available in the location:

Go to the location \Assets and Compliance\Overview\Device Collections\Software Updates\Servers

Collection Name : **Production servers Patching Deployment Only**

The patch deployment process will be the same as mentioned in the patch deployment on test servers



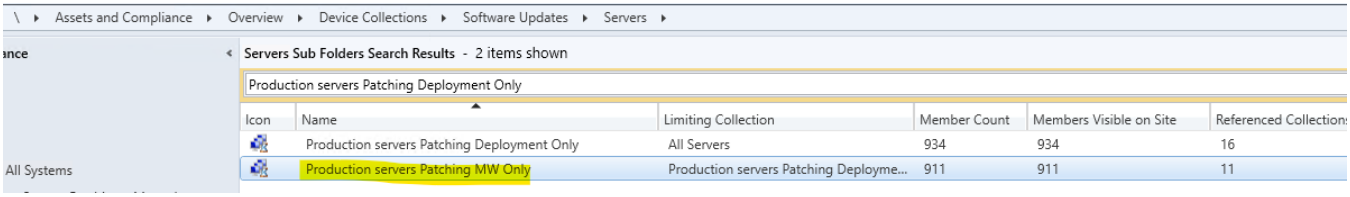
Set MW on the PROD Servers:

Server details available in the location:

Go to the location \Assets and Compliance\Overview\Device Collections\Software Updates\Servers

Collection Name: **Production servers Patching MW Only**

Details were already mentioned above on how to set the MW on collection.



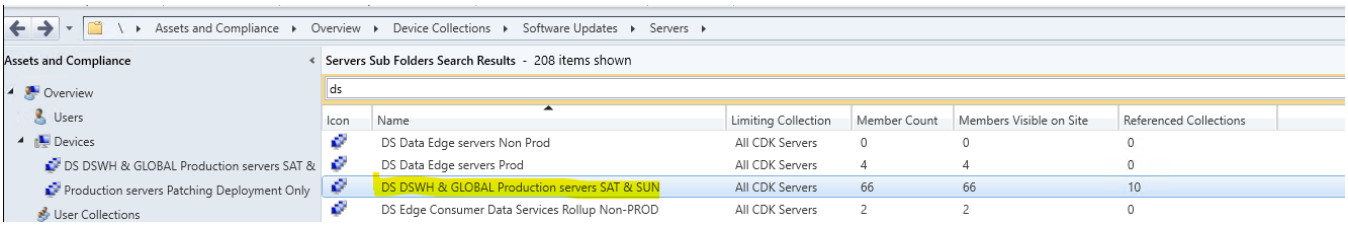
Internal Server PROD Patching:

Server details available in the location:

Go to the location \Assets and Compliance\Overview\Device Collections\Software Updates\Servers

Collection Name: **DS DSWH & GLOBAL Production servers SAT & SUN**

The patch deployment process will be the same as mentioned in the patch deployment on test servers



Set MW on the Internal Server PROD Patching:

Collection Name: **DS DSWH & GLOBAL Production servers SAT & SUN**

The MW should be set as mentioned above